Math 105: Calculus with Algebra
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quiz archive

Gradescope code: B5J8EN

Math 105 Fall 2026 - Take-Home Quiz 1 Solutions

Due Sunday 9/6/2026 - at 11:55 PM

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(1) Solve the equation $|x-2| = 7$ for x.

Solution. Since $|x-2| = 7$, either $x-2 = 7$ or $x-2 = -7$. Thus $x = 9$ or $x = -5$.

(2) Solve the inequality $|x+3| < 4$.

Solution. The inequality $|x+3| < 4$ is equivalent to $-4 < x+3 < 4$. Subtracting 3 throughout gives $-7 < x < 1$.

Thus the solution is $(-7, 1)$. On the number line, shade the interval between -7 and 1, with open circles at -7 and 1.

(3) Express

$$1 + \frac{1}{\frac{x}{x+2}}$$

as a single fraction.

Solution. We have

$$1 + \frac{1}{\frac{x}{x+2}} = 1 + \frac{x+2}{x} = \frac{x+x+2}{x} = \frac{2x+2}{x}.$$

Thus the single fraction is $\dfrac{2x+2}{x}$.

The original expression is defined only when $x \neq -2$ and $x \neq 0$.